

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	20 February 2026
Team ID	LTVIP2026TMIDS24635
Project Name	Flight finder-navigating your air travel options
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

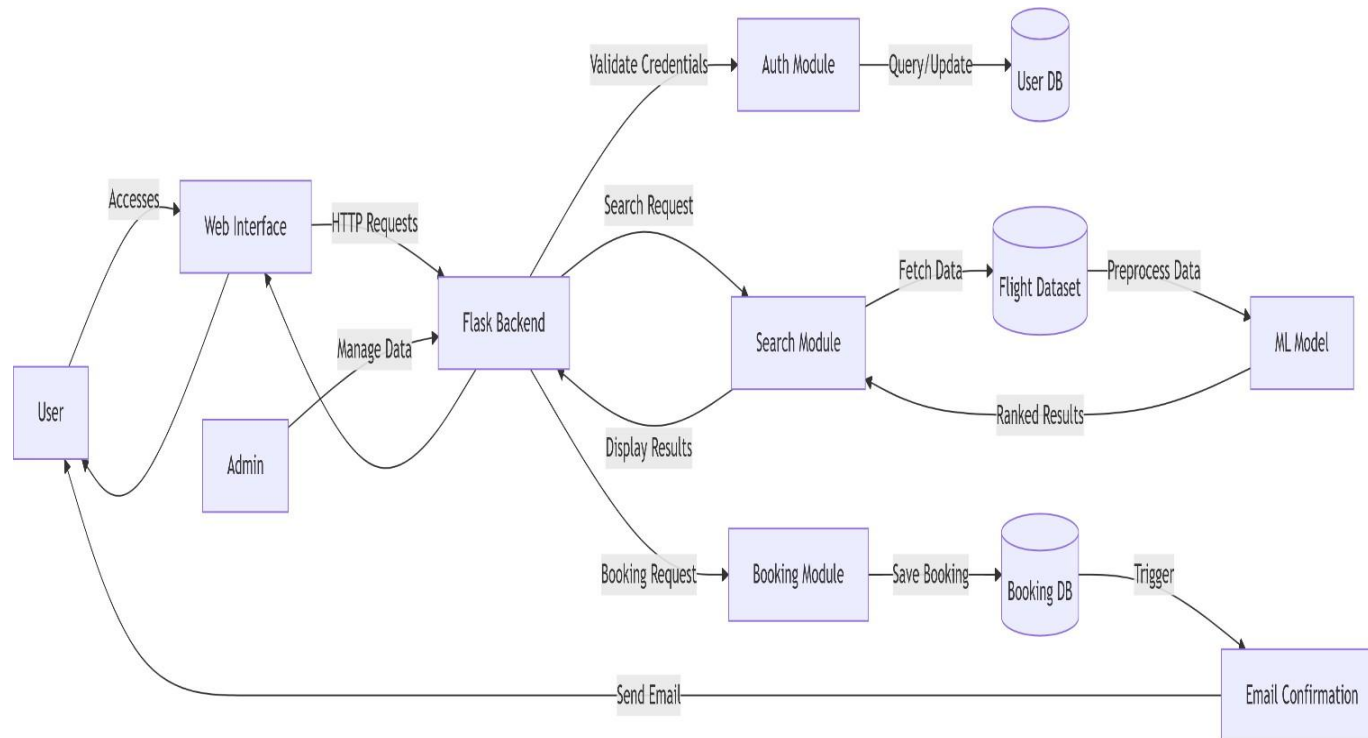


Table 1: System Components and Technology Stack

S.No	Component	Description	Technology
1.	User Interface	Web interface for users, flight agencies, and admin to interact with the system	HTML, CSS, JavaScript, Bootstrap / React JS
2.	Application Logic-1	Backend logic for user registration, login, flight search, booking, and confirmations	Python (Flask) / Java
3.	Application Logic-2	Business logic for flight agency registration, flight addition, and updates	Python (Flask)
4.	Application Logic-3	Admin logic for approving agency accounts and flight listings	Python (Flask)
5.	Authentication & Authorization	User, agency, and admin authentication with role-based access	Flask Authentication / JWT
6.	Database	Stores user data, agency data, flight details, and booking records	MongoDB
7.	File Storage	Storage for flight datasets (CSV/Excel) and logs	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Optional flight data integration (if live data is used)	Public Flight APIs (optional)
9.	External API-2	Email notification service for registration and booking confirmation	SMTP / Gmail API
10.	Machine Learning Model	predict flight pricing or recommend best options	Python, Scikit-learn
11.	Infrastructure (Server / Cloud)	Hosting and deployment of the application	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Uses open-source tools and libraries for development, data processing, and ML	Flask, Pandas, Scikit-learn, Bootstrap
2.	Security Implementations	Ensures secure authentication, authorization, and data protection	Password hashing (bcrypt), JWT, HTTPS, Flask-Login, Email verification
3.	Scalable Architecture	Modular design with clear separation of frontend, backend, and database layers	3-tier architecture, Microservices used
4.	Availability	High availability via cloud hosting	Load balancing (Render, Heroku Dynos, etc.)

S.No	Characteristics	Description	Technology
5.	Performance	Efficient queries, ML model caching, optimized routing	MongoDB indexing, Python caching, CDN (optional)
6.	Reliability	Ensures consistent operation without data loss during failures	Transaction management, exception handling
7.	Maintainability	Easy to update, debug, and extend features such as new APIs or services	Modular Flask structure, MVC pattern
8.	Portability	Application can run on local systems or cloud environments	Localhost, Cloud platforms
9.	Data Management	Efficient storage and retrieval of user, agency, flight, and booking data	MongoDB
10.	Machine Learning Support	Supports price prediction and flight recommendation logic	Python, Scikit-learn